Chapter 1: Numerical Methods

In this chapter, we introduce a locally mass conserving numerical framework. A conforming mixed finite element method (MFEM) is proposed to solve the static Darcy-Stokes coupled methanics system. The proposed conforming MFEM captures the normal fluxes across element edges, ensuring compatibility with local mass conservation. The transport system is solved using a finite volume method (FVM), which inherently ensures local mass conservation.

We begin by introducing the mesh partition used for both the MFEM and the FVM schemes. Based on the mesh partition, two stable element pairs are presented, one for Darcy flow and one for Stokes flow. We then present a noval nonlinear WENO weighting scheme for the FVM. Error estimates are provided for both the MFEM and the FVM schemes.

1.1 Coupled Degenerate Darcy-Stokes System

Now we consider a scaled nondimensionalized degenerate Darcy-Stokes coupled boundary-value problem:

Find two velocity-pressure potential pairs $(\check{\mathbf{v}}_r, \check{q}_l)$ and $(\check{\mathbf{v}}_s, \check{q})$ such that

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1.1}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.2}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{1.3}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.4}$$

and

$$
\begin{aligned}
\check{\mathbf{v}}_r \cdot \boldsymbol{\nu} = g_r \quad & \text{on } \Gamma_{u,D}, \\
\check{\mathbf{v}}_s = \mathbf{g}_s \quad & \text{on } \Gamma_{u,S}, \\
p = p_0 \quad & \text{on } \Gamma_{i,D}, \\
\boldsymbol{\tau} \cdot \boldsymbol{\nu} - p = \boldsymbol{t}_0 \quad & \text{on } \Gamma_{i,S},
\end{aligned}
\tag{1.5}
$$

where $\Gamma_{u,D}$ and $\Gamma_{u,S}$ denote the part of boundary we assign essential boundary conditions for Darcy and Stokes problem respectively, and $\Gamma_{i,D}$ and $\Gamma_{i,S}$ represent the part of boundary we assign natural boundary conditions respectively. We have $\Gamma_{u,D} \cup \Gamma_{i,D} = \partial\Omega$ and $\Gamma_{u,S} \cup \Gamma_{i,S} = \partial\Omega$.

We expand the gradient term in equation (1.1) and the divergence term in equation (1.2) as

$$
\begin{aligned}
\phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) &= \phi^{1/2+\Theta}\nabla\check{q}_l + \phi^{\Theta-1/2}\check{q}_l\nabla\phi, \\
\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) &= \phi^{1/2+\Theta}\nabla\cdot\check{\mathbf{v}}_r + \phi^{\Theta-1/2}\nabla\phi\cdot\check{\mathbf{v}}_r,
\end{aligned}
\tag{1.6}
$$

where they are well-defined provided that

$$
\phi^{\Theta-1/2}\nabla\phi \in (L^\infty(\Omega))^2. \tag{1.7}
$$

A scaled Hilbert space is defined as follows for this specific scaled formulation, where the scaled velocity $\check{\mathbf{v}}_r$ should lie in.

Definition 1.1. Consider a scaled Hilbert space:

$$
\mathbb{V}_D^\phi = H_\phi(\mathrm{div};\Omega) = \Big\{\mathbf{v} \in (L^2(\Omega))^2 : \phi^{-1/2}\nabla\cdot(\phi^{1+\Phi}\mathbf{v}) \in L^2(\Omega)\Big\}, \tag{1.8}
$$

which is a Hilbert space equipped with the inner product

$$
(\mathbf{u},\mathbf{v})_{\mathbb{V}_D^\phi} = (\mathbf{u},\mathbf{v}) + (\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{u}), \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})). \tag{1.9}
$$

The normal trace on $\partial\Omega$ is well-defined as

$$
\left\|\phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu}\right\|_{H^{-1/2}(\partial\Omega)} < C_\Omega\Big\{ \|\mathbf{v}\|_{L^2(\Omega)} + \left\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})\right\|_{L^2(\Omega)} \Big\} \tag{1.10}
$$

subsequently, we have the space $H_\phi^{-1/2}(\mathrm{div};\partial\Omega)$ understood as the image of this normal trace operator.

1.1.1 The weak formulation

We define the following function spaces for the scaled Darcy-Stokes coupled system as

$$
\begin{aligned}
\mathbb{V}_D^{\phi} &= \Big\{ \mathbf{v} \in H_{\phi}(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{v}\cdot\boldsymbol{\nu} = \mathbf{0} \text{ on } \Gamma_{u,D} \Big\},\\
\mathbb{U}_D^{\phi} &= \Big\{ \mathbf{u} \in H_{\phi}(\mathrm{div};\Omega) : \phi^{1/2+\Theta}\mathbf{u}\cdot\boldsymbol{\nu} = \mathbf{g}_r \text{ on } \Gamma_{u,D} \Big\} = \tilde{\mathbf{g}}_r + \mathbb{V}_D^{\phi},\\
\mathbb{W}_D &= L^2(\Omega),\\
\mathbb{V}_S &= \Big\{ \mathbf{v} \in (H^1(\Omega))^2 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S} \Big\},\\
\mathbb{U}_S &= \Big\{ \mathbf{u} \in (H^1(\Omega))^2 : \mathbf{u} = \mathbf{g}_s \text{ on } \Gamma_{u,S} \Big\} = \tilde{\mathbf{g}}_s + \mathbb{V}_S,\\
\mathbb{W}_S &= L^2(\Omega),
\end{aligned}
\tag{1.11}
$$

where each space is equipped with their natural norms. Here, $\tilde{\mathbf{g}}_r$ and $\tilde{\mathbf{g}}_s$ are finite energy lift of Dirichlet data $\mathbf{g}_r$ and $\mathbf{g}_s$ respectively.

We understand the boundary values in the sense of Trace Theorem **??** and the scaled normal trace operator (1.10). We assume regularity for Dirichlet data $\mathbf{g}_r \in (H_{\phi}^{1/2}(\mathrm{div};\Gamma_{u,D}))^2$ and $\mathbf{g}_s \in (H^{1/2}(\Gamma_{u,S}))^2$. We understand $\tilde{\mathbf{g}}_r \in H_{\phi}(\mathrm{div};\Omega)$ and $\tilde{\mathbf{g}}_s \in (H^1(\Omega))^2$ as a continuous extension from the boundary into the domain.

A compatibility condition enforcing mass conservation on the entire domain is needed if essential boundary condition is prescribed on the entire boundary $\partial\Omega$, i.e., when $\Gamma_{u,D} = \Gamma_{u,S} = \partial\Omega$ the Dirichlet data satisfies

$$
\int_{\partial\Omega} (\mathbf{g}_r + \mathbf{g}_s)\cdot\boldsymbol{\nu}\, ds = 0. \tag{1.12}
$$

The standard mixed form for the proposed coupled system reads as follows:

Find $\check{\mathbf{v}}_r \in \mathbb{U}_D^{\phi}$, $\check{q}_l \in \mathbb{W}_D$, $\check{\mathbf{v}}_s \in \mathbb{U}_S$, and $\check{q} \in \mathbb{W}_S$ such that

$$(\check{\boldsymbol{\tau}}(\mathbf{v}_s), \nabla\boldsymbol{\psi}_s) - (\check{q}, \nabla\cdot\boldsymbol{\psi}_s) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_s) \qquad \forall\boldsymbol{\psi}_s \in \mathbb{V}_S, \tag{1.13}$$

$$(\nabla\cdot\check{\mathbf{v}}_s, \omega) + \Big(\frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega\Big) = 0 \qquad \forall\omega \in \mathbb{W}_S, \tag{1.14}$$

$$(\check{\mathbf{v}}_r, \boldsymbol{\psi}_r) - (\check{q}_l, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)) = 0, \qquad \forall\boldsymbol{\psi}_r \in \mathbb{V}_D^{\phi}, \tag{1.15}$$

$$(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \Big(\frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}), \omega_f\Big) = 0 \qquad \forall\omega_f \in \mathbb{W}_D. \tag{1.16}$$

We state well-posedness result of the scaled model as follows.

Lemma 1.1. *(Existence and Uniqueness)*

Assume the condition (1.7) *holds,* $0 \le \phi \le \phi^* < 1$. *There exists a unique solution to the scaled formulation* (1.13)–(1.16), *and it satisfies*

$$\|\check{\mathbf{v}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r)\big\|_{(L^2(\Omega))^2} + \|\check{q}_l\|_{(L^2(\Omega))^2} + \|\check{\mathbf{v}}_s\|_{(H^1(\Omega))^2} + \|\check{q}\|_{(L^2(\Omega))^2}$$
$$\le C\big\{|\rho_r| + \|\tilde{\mathbf{g}}_r\|_{(L^2(\Omega))^2} + \big\|\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\tilde{\mathbf{g}}_r)\big\|_{(L^2(\Omega))^2} + \|\tilde{\mathbf{g}}_s\|_{H^1(\Omega)}\big\}. \tag{1.17}$$

Proof. See Arbogast et al. (2017). □

1.1.2 The scaled mixed finite element method

We discretize the scaled mixed formulation with two space pairs $\mathbb{V}_{D,h}\times\mathbb{W}_{D,h} \subset H(\mathrm{div};\Omega)\times L^2(\Omega)$ and $\mathbb{V}_{S,h}\times\mathbb{W}_{S,h} \subset (H^1(\Omega))^2\times L^2(\Omega)$. Recall the aforementioned inf-sup stable mixed finite element space pairs on $E \in \mathcal{T}_h$:

- For the Darcy part of the system.

$$\begin{aligned} \mathbb{V}_{D,h}(E) &= \mathbb{V}_1^0(E) = (\mathbb{P}_1(E))^2 \oplus \operatorname{curl}\mathbb{S}_2^{\mathcal{DS}}(E) \\ \mathbb{W}_{D,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E) \end{aligned} \tag{1.18}$$

- For the Stokes part of the system.

$$
\begin{aligned}
\mathbb{V}_{S,h}(E) &= \mathbb{V}_1^{\mathcal{BR}}(E) = (\mathcal{DS}_1(E))^2 \oplus \mathrm{span}\{\mathbf{b}_0, \mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\} \\
\mathbb{W}_{S,h}(E) &= \mathbb{W}(E) = \mathbb{P}_0(E).
\end{aligned} \tag{1.19}
$$

We can impose essential boundary conditions by projecting extensions $\tilde{\mathbf{g}}_r$ and $\tilde{\mathbf{g}}_s$ into the finite element spaces as $\hat{\mathbf{g}}_r \in \mathbb{V}_{D,h}$ and $\hat{\mathbf{g}}_s \in \mathbb{V}_{S,h}$.

We also define

$$
\begin{aligned}
\mathbb{V}_{D,h,0} &= \{\mathbf{v} \in \mathbb{V}_{D,h} : \mathbf{v} \cdot \boldsymbol{\nu} = 0 \text{ on } \Gamma_{u,D}\} \\
\mathbb{V}_{S,h,\mathbf{0}} &= \{\mathbf{v} \in \mathbb{V}_{S,h} : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_{u,S}\}
\end{aligned} \tag{1.20}
$$

The scaled mixed finite element method: Find $\check{\mathbf{v}}_{r,h} \in \mathbb{V}_{D,h,0} + \hat{\mathbf{g}}_r$, $\check{q}_{r,h} \in \mathbb{W}_{D,h}$, $\check{\mathbf{v}}_{s,h} \in \mathbf{V}_{S,h,\mathbf{0}} + \hat{\mathbf{g}}_s$ and $\check{q}_h \in \mathbb{W}_{S,h}$ such that

$$
(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}), \nabla\boldsymbol{\psi}_{s,h}) - (\check{q}_h, \nabla\cdot\boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n}, \boldsymbol{\psi}_{s,h}), \qquad \forall \boldsymbol{\psi}_{s,h} \in \mathbb{V}_{S,h,\mathbf{0}}, \tag{1.21}
$$

$$
(\nabla\cdot\check{\mathbf{v}}_{s,h}, \omega_h) - \left(\frac{\phi^{1/2}}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_h\right) = 0, \qquad \forall \omega_h \in \mathbb{W}_{S,h}, \tag{1.22}
$$

$$
(\check{\mathbf{v}}_{r,h}, \boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h}, \phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall \boldsymbol{\psi}_{r,h} \in \mathbb{V}_{D,h,0}, \tag{1.23}
$$

$$
(\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}), \omega_{l,h}) + \left(\frac{1}{1-\phi}(\check{q}_{l,h} - \phi^{1/2}\check{q}_h), \omega_{l,h}\right) = 0, \qquad \forall \omega_{l,h} \in \mathbb{W}_{D,h}. \tag{1.24}
$$

1.1.3 Local Mass Conservation

Before we continue the discussion, we need to make sure that the formulation is well-defined, i.e., not dividing by zero. The term $\phi^{-1/2}$ in the symmetric equations (1.23) and (1.24) causes trouble when there is no melting, i.e., $\phi = 0$. In order to deal with it, we define an element-averaged variable ϕ_E as

$$
\phi_E = \frac{1}{|E|}\int_E \phi(\mathbf{x})\, d\mathbf{x}, \tag{1.25}
$$

where $|E|$ is the area of element E. Then the two terms involving divergence is modified as

$$
\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v})|_E = \begin{cases} \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\mathbf{v}) & \text{if } \overline{\phi}_E \neq 0, \\ 0 & \text{if } \overline{\phi}_E = 0, \end{cases} \tag{1.26}
$$

where the point-wise term $\phi^{-1/2}$ is modified to be a cell-averaged term $\phi_E^{-1/2}$. We expand the integral term involving divergence according to the divergence theorem as

$$\int_E \phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})\omega_{l,h}\,d\mathbf{x} = \int_{\partial E}\phi_E^{-1/2}\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\boldsymbol{\nu}\omega_{l,h}\,ds + \int_E \phi_E^{-1/2}\phi^{1+\Theta}\boldsymbol{\psi}_{r,h}\cdot\nabla\omega_{l,h}\,d\mathbf{x}. \tag{1.27}$$

The second term vanishes when $\omega_{l,h}$ is constant on element E. We now propose our local mass conserved version by replacing the term $\phi^{1/2}$ in equation (1.22) with $\phi_E^{1/2}$ as

$$(\boldsymbol{\tau}(\check{\mathbf{v}}_{s,h}),\nabla\boldsymbol{\psi}_{s,h}) - (\check{q}_h,\nabla\cdot\boldsymbol{\psi}_{s,h}) = -((1-\phi)\mathbf{n},\boldsymbol{\psi}_{s,h}) \qquad \forall\boldsymbol{\psi}_{s,h},\in\mathbb{V}_{S,h,\mathbf{0}}, \tag{1.28}$$

$$(\nabla\cdot\check{\mathbf{v}}_{s,h},\omega_h) - \Big(\frac{\phi\phi_E^{-1/2}}{1-\phi}(\check{q}_{l,h}-\phi_E^{1/2}\check{q}_h),\omega_h\Big) = 0, \qquad \forall\omega_h\in\mathbb{W}_{S,h}, \tag{1.29}$$

$$(\check{\mathbf{v}}_{r,h},\boldsymbol{\psi}_{r,h}) - (\check{q}_{l,h},\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\boldsymbol{\psi}_{r,h})) = 0, \qquad \forall\boldsymbol{\psi}_{r,h}\in\mathbb{V}_{D,h,0}, \tag{1.30}$$

$$(\phi_E^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h}),\omega_{l,h}) + \Big(\frac{\phi\phi_E^{-1}}{1-\phi}(\check{q}_{l,h}-\phi_E^{1/2}\check{q}_h),\omega_{l,h}\Big) = 0, \qquad \forall\omega_{l,h}\in\mathbb{W}_{D,h}. \tag{1.31}$$

To see local mass conservation of the fluid, let $E\in\mathcal{T}_h$ be any element. The equation (1.31) gives

$$\int_E \nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_{r,h})d\mathbf{x} + \int_E \frac{\phi}{1-\phi}(\check{q}_{l,h}-\phi_E^{1/2}\check{q}_h)dx = 0, \tag{1.32}$$

and the equation (1.29) gives

$$\int_E \nabla\cdot\mathbf{v}_{s,h}d\mathbf{x} - \int_E \frac{\phi}{1-\phi}(\check{q}_{l,h}-\phi_E^{1/2}\check{q}_h)dx = 0. \tag{1.33}$$

We define an auxiliary space ω_E being 1 on E and 0 elsewhere. We sum equations (1.29) and (1.31) together assuming that $\omega_h=\omega_E\in\mathbb{W}_{S,h}$ and $\omega_{l,h}=\phi_E^{-1/2}\omega_E\in\mathbb{W}_{D,h}$. The local mass conservation of the phase-averaged mixture is now retrieved as

$$\int_E \nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r+\check{\mathbf{v}}_s)d\mathbf{x} = 0. \tag{1.34}$$

1.1.4 Recovery of the unscaled Darcy variables

The unscaled Darcy potential is $\check{q}_f = \phi^{-1/2}\tilde{\check{q}}_l$, see (**??**). We can reterive an approximate Darcy potential. We assume that $\check{q}_{f,h} \in \mathbb{W}_{D,h}$, so on an element E we define

$$\check{q}_{l,h}|_E = \begin{cases} \phi_E^{-1/2}\tilde{\check{q}}_{l,h}|_E & \text{if} \phi_E \neq 0, \\ 0 & \text{if} \phi_E = 0, \end{cases} \tag{1.35}$$

where the liquid pressure potential $\check{q}_l$ does not hold any physical meanings when there is no melting presents ($\phi = 0$).

The relative Darcy velocity is given by $\check{\mathbf{v}}_r = \phi^{\Theta}\tilde{\check{\mathbf{v}}}$, see (**??**) as well. To maintain local mass balance, we enforce that on each edge $e_i \subset \partial E$

$$\int_{e_i} \phi^{1+\Theta}\tilde{\check{\mathbf{v}}}_r \cdot \boldsymbol{\nu}_i \, ds = \int_{e_i} \phi \check{\mathbf{v}}_{r,h} \cdot \boldsymbol{\nu} \, ds. \tag{1.36}$$

1.1.5 Uniqueness and Convergence properties

Lemma 1.2. *Assuming* (1.7) *holds, there exists a unique solution to the modified local mass conservation modified scaled mixed finite element method* (1.28)–(1.31).

Proof. See Arbogast et al. (2017). The local mass conservation modification replace the term $1/(1-\phi)$ with $(\phi\phi_E^{-1})/(1-\phi)$,which makes it completely analogous to the previous proof. Hence, we are not repeating here. □

We now derive a bound for the local mass conservation modified scaled mixed finite element method.

We derive a bound for the error by taking the difference of

We use the projection operators defined in

1.2 Saddle Point System

Saddle point system is widely spread in technical and science scenarios, such as constrained optimization problem, image reconstruction and optimal control. Saddle point system has been widely and systematically studied, e.g., by Benzi et al. (2005). In our case, saddle point system is formed along with the Mixed Finite Element discretization. In this section, we give a brief overview of properties of a saddle point system and some common ways to solve such system. An abstract saddle point system can be represented as

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{f} \\ \mathbf{g} \end{bmatrix}. \tag{1.37}$$

Assuming the following properties:

- $\mathbf{A}$ is symmetric: $\mathbf{A} = \mathbf{A}^T$;
- $\mathbf{A}$ is nonsingular: $\mathbf{A}^{-1}$ exists;
- $\mathbf{C}$ is symmetric and positive semidefinite;
- $\mathbf{B}$ is rank deficient.

1.2.1 Reduced Linear System

Now we discuss the saddle point system (SPS) induced by MFEM method in details including the boundary conditions. To begin with, we regroup dofs according to the boundary condition.

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_K & \mathbf{A}_M \\ \mathbf{A}_M^T & \mathbf{M} \end{bmatrix}, \tag{1.38}$$

where $\mathbf{A}_K$ consists of interaction within interior dofs and natural boundary condition dofs, and $\mathbf{A}_M$ consists of iteraction between interior dofs, natural boundary dofs and essential boundary dofs.

We perform the same treatment for matrix $\mathbf{B}$ and right hand side vector $\mathbf{f}$, by regrouping dofs according to its iteration with boundary conditions.

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_K \\ \mathbf{B}_M \end{bmatrix}, \text{and} \quad \mathbf{F} = \begin{bmatrix} \mathbf{F}_K \\ \mathbf{F}_M \end{bmatrix} \tag{1.39}$$

We rewrite the linear system with the regrouped dofs and form a reduced system as

$$\begin{bmatrix} \mathbf{A}_K & \mathbf{B}_K \\ \mathbf{B}_K^T & -\mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_K - \mathbf{A}_M \mathbf{g} - \mathbf{g}_N \\ \mathbf{G} - \mathbf{B}_M^T \mathbf{g} \end{bmatrix}, \tag{1.40}$$

where $\mathbf{g}$ denotes the values prescribed for essential boundary, and $\mathbf{g}_N$ is the vector generated by the natural boundary condition part. The reformed reduced system is still a saddle point syste.

Now we take a closer look at the formed linear system. The no melting (zero porosity $\phi = 0$) region adds difficulties to solving the formed saddle point system. On one hand, the existence of zero porosity region guarantees that $\mathbf{B}$ matrix formed in Darcy system does not has full column rank. Therefore the schur complement

$$\mathbf{S} = -(\mathbf{C} + \mathbf{B}\mathbf{A}^{-1}\mathbf{B}^T),$$

is usually not invertable directly. On the other hand, in addition to the constant null space induced by gradient of pressure, the null space of $\mathbf{B}$ matrix $\ker(\mathbf{B})$ in our darcy problem is not fixed due to evloution of melting (change of zero porosity region).

Taking into consideration that the solver will be implemented in parallel, a iterative solver is more suitable than direct solver.

1.2.2 Schur Complement

To begin with, we discuss a "direct" way to solve this derived saddle point system. Now consider a general saddle point system (1.37) with symmetric positive-definite matrix $\mathbf{A}$ as

$$\begin{aligned} \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{y} &= \mathbf{f}, \\ \mathbf{B}^T\mathbf{x} - \mathbf{C}\mathbf{y} &= \mathbf{g}. \end{aligned} \tag{1.41}$$

We represent the solution vector $\mathbf{x}$ as

$$\mathbf{x} = \mathbf{A}^{-1}(\mathbf{f} - \mathbf{B}\mathbf{y}), \tag{1.42}$$

where $\mathbf{A}^{-1}$ is computed with conjugate gradient algorithm. We then eliminate $\mathbf{x}$ to represent $\mathbf{y}$ as

$$\mathbf{y} = \mathbf{S}^{-1}(\mathbf{g} - \mathbf{B}^T\mathbf{A}^{-1}\mathbf{f}), \tag{1.43}$$

where $\mathbf{S}^{-1}$ is computed with Minimal Resigual algorithm Paige and Saunders (1975) due to the formed Schur complement being a symmtric indefinite system.

1.2.3 Uzawa Iteration

Uzawa algorithm was originally introduced by K.J.Arrow et al. (1958), which gained popularity in computational fluid dynamics. It has later been applied to the case of discretizing Stokes problem with Mixed Finite Element. Some inexact inner solver to accelerate uzawa iteration was discussed by Elman and Golub (1994) and Bramble et al. (1997).

We start by stating an abstract inexact ezawa iteration algorithm in Algorithm 1. Some choose diagonal matrix $\alpha\mathbf{I}$ for precondictioner $\mathbf{Q}_A$. However, such

Algorithm 1 Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.
2: **while** res $>$ tol **do**
3: Obtain $\tilde{\mathbf{x}} = \mathbf{Q}_A^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$
5: Obtain $\tilde{\mathbf{y}} = \mathbf{Q}_B^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$
6: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
7: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$
8: **end while**

simple matrix wouldn't generate a convergence result when applied to our coupled problem. In the next section, we introduce a modified algorithm that would solve our system.

1.2.4 Modified Uzawa Iteration

The linear system formed from the equations (1.28), (1.29), (1.30) and (1.31) is

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{1.44}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{1.45}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{1.46}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{1.47}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system.

Algorithm 2 Modified Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$, $\mathbf{y}_0 = \mathbf{0}$, tolence tol $= 1$ and residual $res = 0$.

2: **while** res $>$ tol **do**

3: Obtain $\tilde{\mathbf{x}} = \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$

4: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \tilde{\mathbf{x}}$

5: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$

6: Obtain $\tilde{\mathbf{y}} = \text{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$

7: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

8: Update res $= \|\tilde{\mathbf{x}}\|_2 + \|\tilde{\mathbf{y}}\|_2$

9: **end while**

1.2.5 Numerical Test

We test our MFEM setup with two different scenarios. First, we test a pseudo 1D problem but with 2D elements equipped with three different porosity scenarios. Second, we test a more complex yet a more realistic problem representing Mid Ocean Ridge where no true solution is presented.

Problem Setup 1: Column Compaction.

In this problemsetup, we perform a convergence test on a 2D grid with a pseudo 1D problem. We study a compacting column in one direction. The column extends over $y \in [-L, L]$ and has no flow through top and bottom boundaries. We prescribe the following boundary conditions

- Bottom Edge;
  - Essential boundary condition:
    $$\begin{cases} \check{\mathbf{v}}_s = \mathbf{0}; \\ \mathbf{b} = \mathbf{0}; \end{cases}$$
  - Essential boundary condition: Normal Darcy flux $\check{\mathbf{v}}_r \cdot \boldsymbol{\nu} = 0$;
- Left and Right side Edges;
  - y component of Stokes velocity is free, x component is kept as zero;
  - Darcy velocity, which is understood as flux is also kept zero;
  - normal flux dof fixed with a zero flux.
- Top Edge;
  - Stokes velocity $\check{\mathbf{v}}_s = \mathbf{0}$;
  - Darcy velocity $\check{\mathbf{v}}_l = \mathbf{0}$;
  - Normal flux is also determined to be zero.

We assume no velocity in x direction, $\check{\mathbf{v}}_s \cdot \mathbf{i} = 0$, and $\check{\mathbf{v}}_r \cdot \mathbf{i} = 0$, where $\mathbf{i}$ is the unit vector in x direction pointing from left to right. Let $u = \phi \check{\mathbf{v}}_r \cdot \mathbf{j}$ and $v = \check{\mathbf{v}}_s \cdot \mathbf{j}$, where $\mathbf{j}$ is the unit vector in y direction pointing upwards.

$$\begin{aligned} u + \phi^2 q_l' &= 0, \\ u' + \phi(q_l - q_s) &= 0, \\ [q_s - \frac{1}{3}(1-4\phi)v']' &= 1 - \phi, \\ v' - \phi(q_f - q_s) &= 0, \end{aligned} \tag{1.48}$$

When no melting presents $\phi = 0$, the equations reduce to

$$u = -v = 0, \quad \text{and} \quad q_s' = 1, \quad \text{and} \quad q_l = 0. \tag{1.49}$$

When melting presents $\phi > 0$, the equations can be reduced to

$$\phi^{2+2\Theta}\left(\frac{3+\phi-4\phi^2}{3\phi}u'\right)' - u = \phi^{2+2\Theta}(1-\phi), \tag{1.50}$$

where all other quantities can be determined according to u as

$$v = -u, \quad \text{and} \quad q_l' = -\phi^{-2-2\Theta}u, \quad \text{and} \quad qs = u'/\phi + q_l. \tag{1.51}$$

When ϕ is constant, we define an auxiliary function R as

$$R = \left(\phi^{2+2\Theta}\frac{3+\phi-4\phi^2}{3\phi}\right)^{-1/2}. \tag{1.52}$$

We test our MFEM code on three porosity settings, and use the closed form solutions derived in Arbogast et al. (2017).

- Constant porosity;

$$\phi_0(z) = \phi_0; \tag{1.53}$$

Closed form solution:

$$\begin{aligned} u &= -\phi_0^2(1-\phi_0)\left[1 - \frac{\cosh(Rz)}{\cosh(RL)}\right], \\ v &= -u, \\ q_l &= (1-\phi_0)\left[z - \frac{1}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right] + c, \\ q_s &= (1-\phi_0)\left[z - \frac{1-4\phi_0}{3+\phi_0-4\phi_0^2}\frac{\phi_0}{R}\frac{\sinh(Rz)}{\cosh(RL)}\right] + c, \end{aligned} \tag{1.54}$$

where c is a constant kernel.

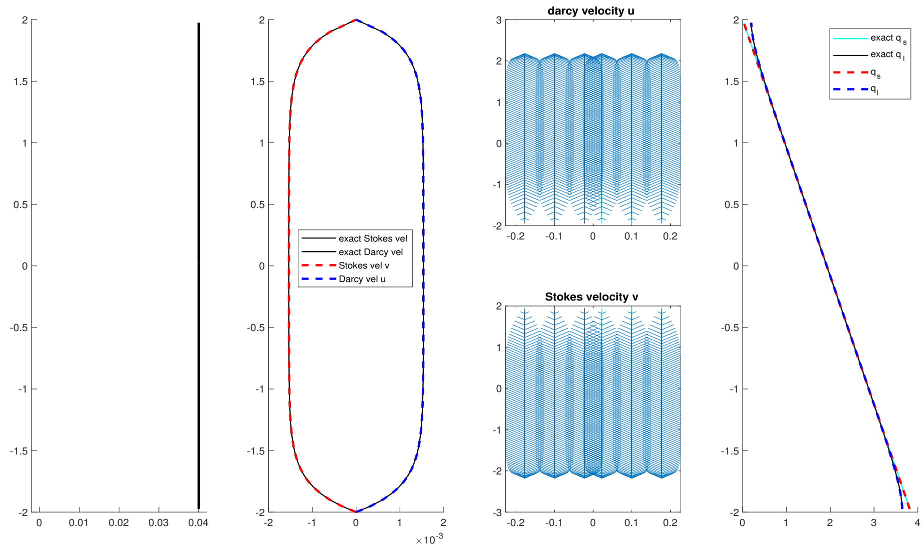

Figure 1.1: Constant porosity compaction column test.

	u		v		q_l		q_s	
$m \times n$	L^2 error	rate	L^2 error	rate	L^2 error	rate	L^2 error	rate
2×20	3.100e-05	-	3.100e-05	-	3.285e-03	-	3.638e-05	-
2×40	8.047e-06	1.95	8.047e-06	1.95	8.731e-04	1.91	9.671e-06	1.91
2×80	2.031e-06	1.99	2.031e-06	1.99	2.218e-04	1.98	2.457e-06	1.98
2×160	5.091e-07	2.00	5.091e-07	2.00	5.567e-05	1.99	6.166e-07	1.99

Table 1.1: Constant porosity compaction column test.

- Piecewise constant porosity;

$$\phi_J(z) = \begin{cases} 0 & \text{if } z > 0, \\ \phi_0 & \text{if } z \leq 0, \end{cases} \tag{1.55}$$

Closed form solution:

We conclude that $u = -v = 0$ when $z > 0$, and $q_l = 0$, $q_s = z + c$. When $z < 0$,

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	2.980e-05	-	2.980e-05	-	3.794e-03	-	4.202e-05	-
2×40	7.681e-06	1.96	7.681e-06	1.96	1.005e-03	1.92	1.114e-05	1.92
2×80	1.958e-06	1.97	1.958e-06	1.97	2.472e-04	2.02	2.738e-06	2.02
2×160	4.974e-07	1.98	4.974e-07	1.98	5.972e-05	2.04	6.615e-07	2.05

Table 1.2: Piecewise Constant porosity compaction column test.

$\phi_0 \neq 0$,

$$\begin{aligned}
u &= \phi_0^2(1-\phi_0)\left[1 - \cosh(Rz) + \frac{1-\cosh(RL)}{\sinh(RL)}\sinh(Rz)\right],\\
v &= -u,\\
q_l &= -(1-\phi_0)\left[z - \frac{1}{R}\sinh(Rz) + \frac{1-\cosh(RL)}{R\sinh(RL)}\cosh(Rz)\right] + c,\\
q_s &= q_l + \frac{u'}{\phi_0} + c,
\end{aligned} \tag{1.56}$$

where c is a constant kernel shifting the pressure potential.

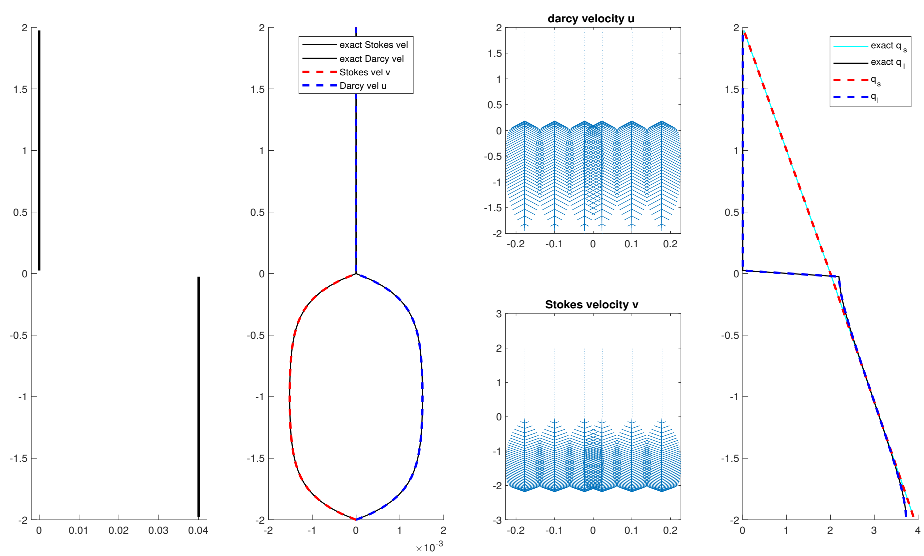

Figure 1.2: Piecewise constant porosity compaction column test.

- Quadratic porosity.

$$\phi_2(z) = \begin{cases} 0 & \text{if } z > 0, \\ \phi_0 z^2 & \text{if } z \leq 0, \end{cases} \tag{1.57}$$

Closed form solution:

We conclude that $u = -v = 0$ when $z > 0$.

$$\begin{aligned} u &= \frac{\phi_0^2}{1-4\phi_0}\left(L^{4-r_1} z^{r_1} - z^4\right), \\ v &= -v \\ q_l &= \frac{1}{1-4\phi_0}\left(z + \frac{L^{4-r_1}(-z)^{r_1-3}}{r_1-3}\right) + c, \\ q_s &= z + c, \end{aligned} \tag{1.58}$$

where

$$r_1 = \frac{3+\sqrt{9+4/\phi_0}}{2}. \tag{1.59}$$

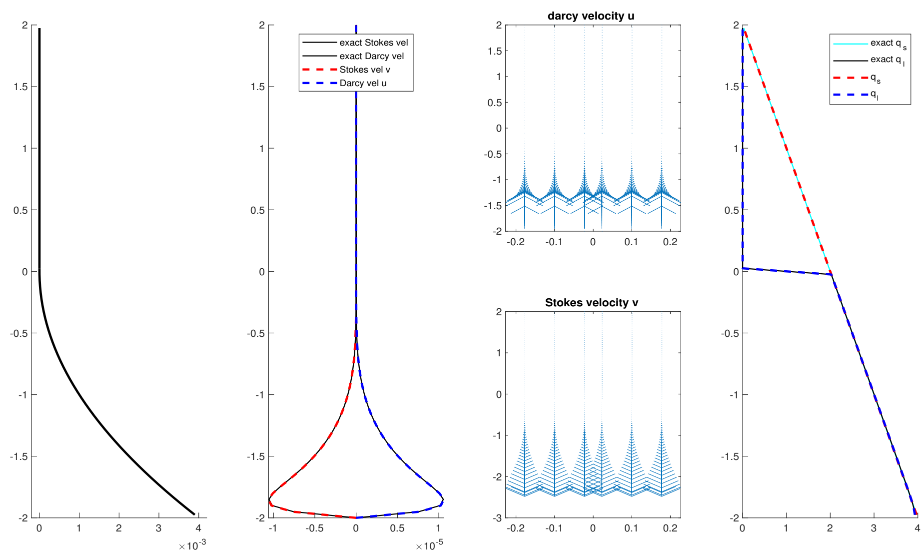

Figure 1.3: Quadratic porosity compaction column test.

Problem Setup 2: Mid-Ocean Ridge.

We now reconsider the Mid-Ocean ridge test case in Arbogast et al. (2017). We solve the full system of equations on a dimensionless rectangular domain $x \in$

$m \times n$	u L^2 error	rate	v L^2 error	rate	q_l L^2 error	rate	q_s L^2 error	rate
2×20	9.616e-07	-	9.616e-07	-				
2×40	3.212e-07	1.58	3.212e-07	1.58				
2×80	9.028e-08	1.83	9.028e-08	1.83				
2×160	2.746e-08	1.72	2.746e-08	1.72				

Table 1.3: Quadratic porosity compaction column test.

$[-0.5, 0.5]$ and $z \in [-0.5, 0.0]$. The boundary conditions are imposed by assigning classical corner flow solution of Stokes velocity in Spiegelman and McKenzie (1987) as essential boundary condition all around the computational domain

$$\mathbf{v}_s = \frac{2U_0}{\pi(x^2+y^2)} \begin{pmatrix} \tan^{-1}(\frac{x}{-y})(x^2+y^2) + xy \\ z^2 \end{pmatrix}, \tag{1.60}$$

where $U_0 = 10^{-9}$m/s ≈ 3.2cm/yr is the constant upwelling velocity of mantle. According to mass conservation condition, we set no Darcy flux across the boundary. We then set a not identically vanishing porosity by the formula

$$\phi(x,z) = \begin{cases} 0.05(\frac{0.375+y}{0.375})^2(1-\frac{|x|}{|y|+l}), & \text{if } |y| \leq 0.375 \text{ and } |x| \leq z+l, \\ 0, & \text{otherwise,} \end{cases} \tag{1.61}$$

where $l = 20\,\mathrm{m}$ is the offset set to avoid singularity computation at the center $x = 0$ we translate the coordinates x to $x - l$ if $x < 0$, and $x + l$ when $x > 0$. In Figure 1.4, we show the porosity distribution in the computational domain. Unlike the previous test conducted in the quarter plane where $x \in (0, 0.5]$, $y \in [-0.5, 0.0]$. We perform the numerical test within the entire plane instead.

In Figure 1.5 and Figure 1.6, we exhibit the nondimensional Stokes velocity $(\check{\mathbf{v}}_s, \check{q}_s)$ and Darcy velocity $(\phi\check{\mathbf{v}}_r, \check{q}_l)$ respectively. We observe the molten magma rise to the surface and focus to the region where most porosity presents. The solid matrix upwell from the bottom of the domain. We observe compaction at degree of melting (some porosity) where the buoyancy can no longer support the solid matrix on top of it due to density difference between solid and liquid. Despite of the presence of large no melting region, the system is solved successfully with modified Uzawa iteration

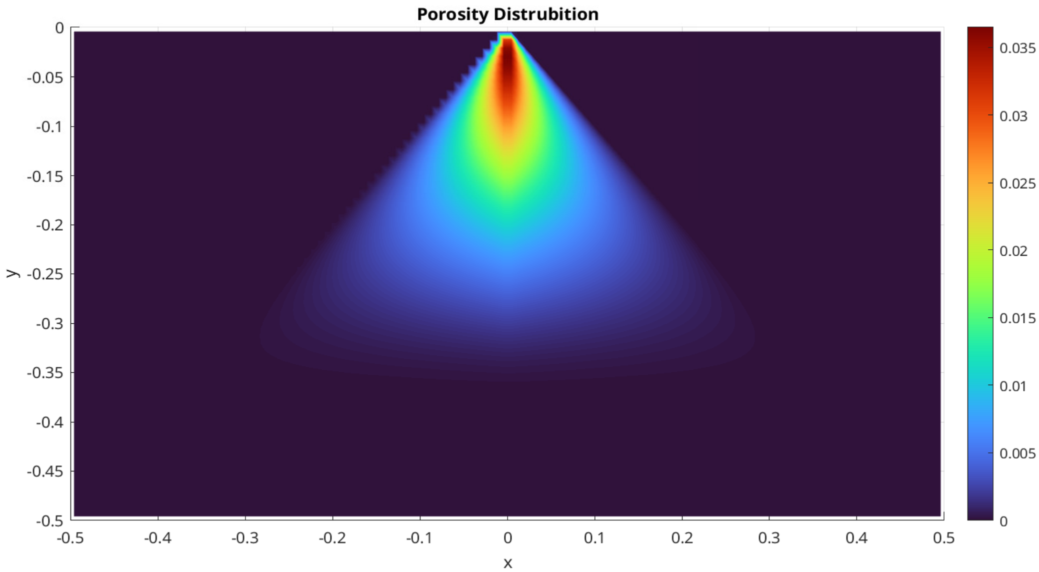

Figure 1.4: Porosity distribution.

1.3 Hyperbolic Transport System

1.4 Finite Volume Method

The finite volume method (FVM) is a discretization method widely used in various scenarios that conservation laws are involved. The finite volume is a robust algorithm that can be applied to arbitriary complex geometry. The most important aspect of FVM that benefits this simulation is that the local mass conservation is achieved naturally. To begin with, we consider a general conservation law retarding a scalar variable u as

$$\begin{cases} \dfrac{\partial u(\mathbf{x},t)}{\partial t} + \nabla \cdot \mathbf{f}(u;\mathbf{x},t) = 0, & t > 0, \mathbf{x} \in \Omega, \\ u(\mathbf{x},0) = u_0(\mathbf{x}), & \mathbf{x} \in \Omega, \end{cases} \tag{1.62}$$

where we assume the domain $\Omega \subset \mathbb{R}^2$. The term $\mathbf{f}$ is a flux transporting scalar variable u. We write the basic semi-discretized form of basic finite volume method as

$$\frac{\partial \overline{u}}{\partial t} + \frac{1}{|E|} \int_{\partial E} \mathbf{F}(\mathbf{x},t) \cdot \boldsymbol{\nu} d\mathbf{x} = 0, \tag{1.63}$$

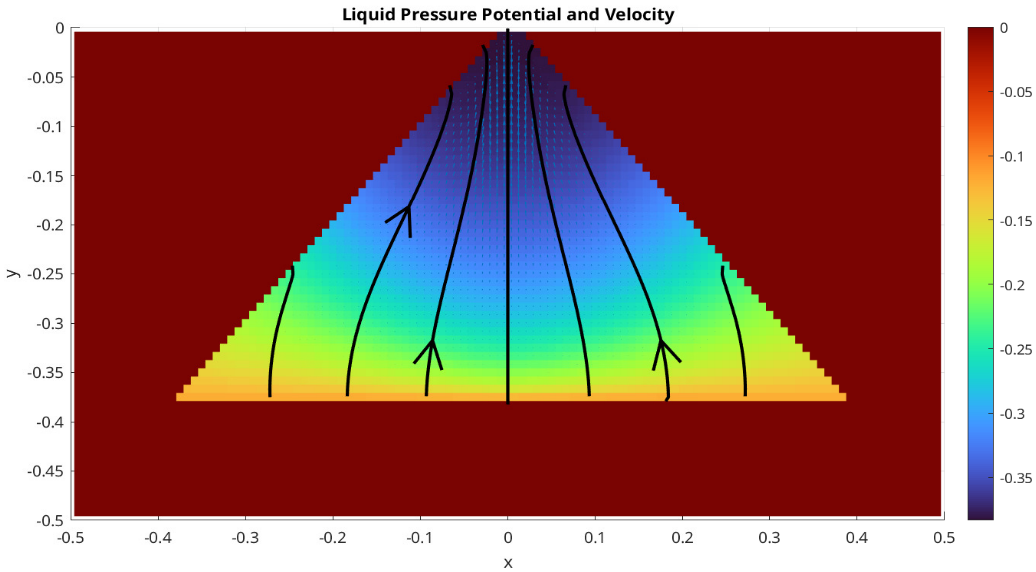

Figure 1.5: Liquid pressure potential and velocity.

where we use the cell-averaged value $\overline{u} = \frac{1}{|E|}\int_E u(\mathbf{x},t)d\mathbf{x}$, and $F(\mathbf{x},t)$ is the approximated numerical flux. We use the Lax-Friedrichs numerical flux

$$\mathbf{F}(u^+,u^-) = \frac{1}{2}[(\mathbf{f}(u^+)+\mathbf{f}(u^-))\cdot\boldsymbol{\nu} - \alpha_{LF}(u^+-u^-)], \tag{1.64}$$

where u^+ and u^- represent values approximated from element $E^+ \in \mathfrak{T}_h$ and $E^- \in \mathfrak{T}_h$ respectively, and $\boldsymbol{\nu}$ is the unit vector pointing from E^+ to E^- orthogonal to the edge. Here, α_{LF} is the Lax-Fredrich stabilizer factor. We have two different ways to compute α_{LF} depending on the knowledge of solution.

- The global LF Stabilizer: $\alpha_{LF} = \max_u|\partial(\mathbf{f}\cdot\boldsymbol{\nu})/\partial u|$;
- The local LF Stabilizer: $\alpha_{LF} = \max(|\partial(\mathbf{f}^+\cdot\boldsymbol{\nu})/\partial u^+|, |\partial(\mathbf{f}^-\cdot\boldsymbol{\nu})/\partial u^-|)$.

The numerical flux $\mathbf{F}$ (1.64) is consistent and conservtive across the edge. We now replace the u^+ and u^- with reconstructed values and rewrite the numerical flux as

$$\mathbf{F}(\mathcal{R}^+,\mathcal{R}-) = \frac{1}{2}[(\mathbf{f}(\mathcal{R}^+)+\mathbf{f}(\mathcal{R}^-)\cdot\boldsymbol{\nu} - \alpha_{LF}(\mathcal{R}^+-\mathcal{R}^-)], \tag{1.65}$$

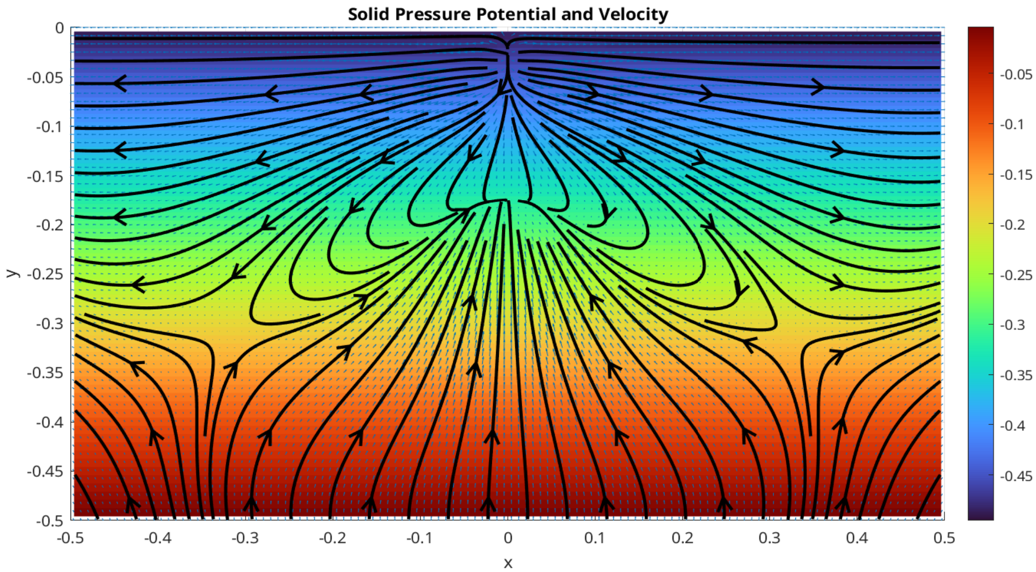

Figure 1.6: Solid pressure potential and velocity.

where $\mathcal{R}$ is a higher order reconstruction value. We evaluate the integrals over each edge $e_i \in \partial E$ with a quadrature rule. We represent length of edge e_i and area of cell E with $|e_i|$ and $|E|$ respectively. Assuming a quadrature rule with G points on each edge, we have

$$\overline{u}_t + \sum_{i=0}^{4} \frac{|e_i|}{|E|} \sum_{g} \omega_{i,g} \mathbf{F}(\mathcal{R}^+_{i,g}, \mathcal{R}^-_{i,g}) = 0, \tag{1.66}$$

where $\omega_{i,g}$ represents gth quadrature points on edge e_i.

1.5 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for reconstruction of discontinuous values with cell averaged values in Arbogast et al. (2024). The ML-WENO reconstruction will be primarily used in reconstruction of values on both sided of the edge for numerical flux computation appears in FVM discretization. It can also be applied to general reconstruction of discontinuous variables, e.g., discontinuous darcy pressure appears on the zero-nonzero porosity interface.

1.5.1 Stencil Polynomial

Given a stencil $\mathcal{S} = \{E_0, E_2, \dots, E_k\}$ of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$, where s and t equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p<s,q<t} c_{p,q} \Big(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p \Big(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{1.67}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is the center point of the stencil $\Omega_{\mathcal{S}} = \cup_{E\in\mathcal{S}} \subset \mathbb{R}^2$, and $h_{\mathcal{S}} = \sqrt{\sum_k E_k/k}$ is the diameter of average cell size. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ be the normalized varaibles, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p<s,q<t} c_{p,q} \xi^p \eta^q. \tag{1.68}$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\overline{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} = \frac{1}{|E|} \int_E u(x, y) dx dy. \tag{1.69}$$

We can calculate coefficents $c_{p,q}$ by imposing a constraint that cell averaged value (1.69) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|} \int_{E_k} Q(\mathbf{x}) d\mathbf{x} = \overline{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \tag{1.70}$$

We will replace $\overline{u}_{E_k}$ with $\overline{u}_k$ in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials $Q_{s,t} = \sum_k \overline{u}_k \phi^k_{s,t}$, where basis polynomials $\phi^k_{s,t}$ can be calculated as

$$\frac{1}{|E_{k'}|} \int_{E_{k'}} \phi^k_{s,t}(\mathbf{x}) d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|} \iint_{\hat{E}_{k'}} \hat{\phi}^k_{s,t}(\xi, \eta) d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall E_{k'}, E_k \in \mathcal{S}, \tag{1.71}$$

which would lead to a $(s \times t) \times (s \times t)$ non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi, \eta) = \sum_k \overline{u}_k \hat{\phi}^k_{s,t}(\xi, \eta). \tag{1.72}$$

We define a linear projection operator $\pi_{\mathcal{S}}$ and let $\pi_{\mathcal{S}} u_{\mathcal{S}} = Q$ be a linear projection onto polynomial space W_p such that $\mathbb{Q}_{s,t} \subset W_p$. We estimate interpolation error with the Bramble-Hilbert Lemma Dupont and Scott (1980) and Bramble and Hilbert (1970) in $H_r(\Omega)$ space. For simplicity, we assume $r = s = t$.

Lemma 1.3. *Let Ω_S be the domain covered by stencil $\mathcal{S}$ in $\mathbb{R}^2$ and let $u \in H^r(\Omega_{\mathcal{S}})$. We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|^2_{H^r(\Omega_S)}. \tag{1.73}$$

Proof.

$$\|u - \pi_S u\| = \tag{1.74}$$

□

In a general 2D situtation when $s \neq t$, the exececssive order will be "wasted", since the order or accuracy will be $\min(s, t)$. However, in some cases, where we aquire the prior knowledge of the solution being qusi-1D, e.g., the transport velocity is fixed in one direction. We can have uneven stencils that has $\max(s, t)$ order of accuracy but in the desired direction. In other cases when derivative is involved, we also need an stretched stencil to compensate for the order loss when approximates derivative.

1.5.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from Jiang and Shu (1996) is defined in analogue to a cell-averaged Sobolev seminorm as:

$$\sigma_{JS} = \sum_{1 \leq |\alpha| \leq m} \int \int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x, y))^2 dx dy, \tag{1.75}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|}/\partial x^{\alpha_1}\partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$
\begin{aligned}
\sigma_{JS} &= \sum_{1\le|\alpha|\le m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dxdy \\
&= \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta)\big)^2 d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'.j'} \sum_{1\le|\alpha|\le m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \sum_{ij}\sum_{i'j'} \overline{u}_{ij}\overline{u}_{i'j'}\sigma_{ij}^{i'j'},
\end{aligned}
\tag{1.76}
$$

where $\sigma_{ij}^{i'j'}$ can be precomputed after mesh is established after mesh is established.

Alternatively, we can expand the Jiang-Shu smoothness indicator according to polynomial coefficients, which paves the way for further simplification. Recall the tensor product polynomial version (1.68),

$$
\begin{aligned}
\sigma_{JS} &= \sum_{1\le|\alpha|m} \\
&= \sum\sum \eta_i c_i c_j
\end{aligned}
\tag{1.77}
$$

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$
\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(1) & \text{Not Smooth,} \end{cases}
\tag{1.78}
$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

1.5.3 Two alternative smoothness indicator

We proposed two alternative ways of defining smoothness indicator in Huang et al. (2023) and Arbogast et al. (2025).

- *Simple smoothness indicator*

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{1.79}$$

where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the computational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

- *Polynomial smoothness indicator*

$$\sigma_P = \sum \eta_i c_i^2 \tag{1.80}$$

We obtain this polynomial smoothness indicator by dropping all the cross terms.

1.5.4 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\{\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L, L \geq 0\}$. The reconstruction of u on target cell $E_{i,j}$ is a convex combination of stencil polynomials (1.72)

$$\mathcal{R}(\mathbf{x}) = \sum_l \tilde{\omega}_l Q_l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{1.81}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{1.82}$$

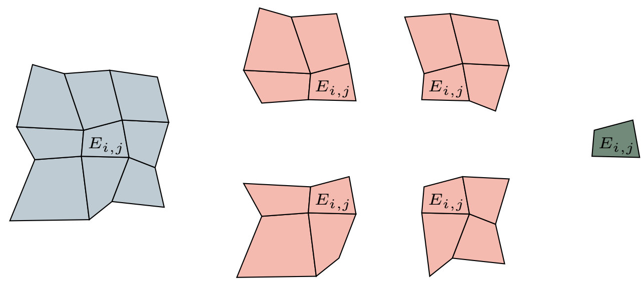

Figure 1.7: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

where $r_l = \max\{s, t\}$, $\epsilon = 1e-4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{1.83}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{1.84}$$

1.5.5 The ML-WENO Reconstruction Error

In this section, we consider the accuracy of our multi-level weighting scheme. We group the set of indices $l \in \mathcal{L} = \{0, 1, 2, \ldots L\}$ according to the order of reconstruction polynomials, i.e., the shape of individual stencil. We denote the set of indices of stencils have s cells in x direction and t cells in y direction as $\mathcal{L}_{s,t}$. We make the assumption that the mesh resolution is fine enough so that there exist at least one stencil that is smooth if shock presents. We define two sets holding indices of stencils with or without jump as

$$\begin{aligned} \text{Smooth} &= \{l : \ u \text{ is smooth on } \mathcal{S}_l, \} \\ \text{Jump} &= \{l : \ u \text{ has a jump on } \mathcal{S}_l\} \end{aligned} \tag{1.85}$$

where we assume Smooth $\cup$ Jump $= \mathcal{L}$ and Smooth $\cap$ Jump $= \emptyset$. We represent the set of indices of stencils of shape (s,t) that is smooth as $\mathcal{L}_{s,t} \cap$ Smooth, and the set of stencils of shat (s,t) that has a jump as $\mathcal{L}_{s,t} \cap$ Jump. For convience, we assume square stencils, where $r = s = t$. The accuracy result can be applied to $s \neq t$ situation directly in the selected direction. We show that the reconstruction is accurate with respect to the target cell E_0 to order

$$r_{\max} = \max\{r : \text{Smooth} \cap \mathcal{L}_{r,r} \neq \emptyset\}. \tag{1.86}$$

In Figure 1.8, we illustrate a (3,2,1) combination of stencils with thow shock presents. Here, Smooth $= \{0, 1\}$, which indicates that stencil $\mathcal{S}_0$ and $\mathcal{S}_1$ do not have jump present. In this case, we expect the accuracy of reconstruction is $r_{\max} = \text{Smooth} \cap \mathcal{L}_{2,2} = 2$.

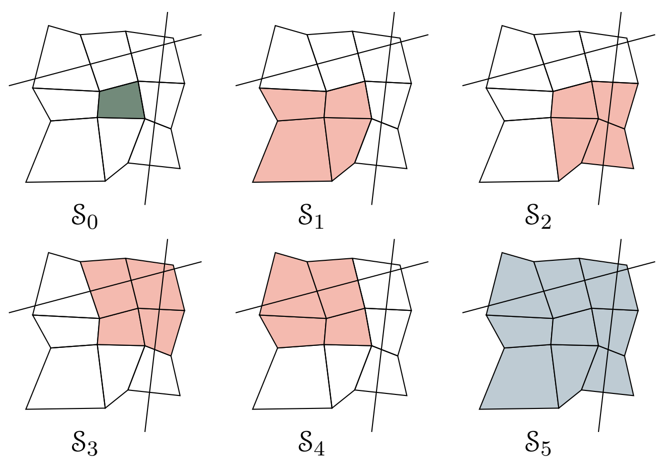

Figure 1.8: An illustration of a (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh with two shock presents.

We state the estimation of reconstruction error show in Arbogast et al. (2024). We start by estimating the error of our nonlinear scaling (1.82) asscoiated to some stencil $\mathcal{S}_l$ for some $m > 0$.

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m}. \tag{1.87}$$

We estimate the reconstruction error for two situtaions.

- $l \in$ Smooth, the smoothness indicator $\sigma_l = Ch^2 + \mathcal{O}(h^3)$;
- $l \in$ Jump, the smoothness indicator $\sigma_l = \Theta(1)$.

Use the asymptotic estimation result from Olver (1997). Let $n \geq m$, $a > 0$, and $b > 0$

$$\begin{aligned} \frac{a}{bh^m + \mathcal{O}(h^n)} &= \frac{ah^{-m}}{b}\big(1 + \mathcal{O}(h^{n-m})\big), \\ \frac{a}{bh^{-n} + \mathcal{O}(h^{-m})} &= \frac{ah^n}{b}\big(1 + \mathcal{O}(h^{n-m})\big). \end{aligned} \tag{1.88}$$

For the smooth stencil $\mathcal{S}_l \in$ Smooth

$$\begin{aligned} \hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^m} &= \frac{\omega_l}{((C+\epsilon)h_0^2 + \mathcal{O}(h^3))^m} \\ &= \omega_l \Big(\frac{h_0^{-2}}{C+\epsilon}(1 + \mathcal{O}(h)) \Big)^m \\ &= \frac{\omega_l}{(D+\epsilon)^m} h_0^{-2m}(1 + \mathcal{O}(h)). \end{aligned} \tag{1.89}$$

In conclusion, we estimate the nonlinear scaling as

$$\hat{\omega}_l = \begin{cases} \dfrac{\omega_l}{(C+\epsilon)^m} h_0^{-2m}(1 + \mathcal{O}(h)), & \text{if} \quad l \in \text{Smooth}, \\ \Theta(1), & \text{if} \quad l \in \text{Jump}. \end{cases} \tag{1.90}$$

Then we estimate the error for the normalized nonlinear weighting (1.84). We expand the sum of nonlinear scaling as

$$\sum_l \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l + \sum_{l \in \text{Jump}} \hat{\omega}_l. \tag{1.91}$$

We estimate the smooth part of the nonlinear weighting as

$$\sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \hat{\omega}_l = \sum_r \sum_{\substack{l \in \\ \text{Smooth} \cap \mathcal{L}_{r,r}}} \frac{\omega_l}{(C+\epsilon)^m} h_0^{-2(ar+\eta)}(1 + \mathcal{O}(h)). \tag{1.92}$$

Assume $h < 1$, we keep the largest $ar + \eta(r)$ number as our estimator

$$\begin{aligned}\sum_r \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \hat{\omega}_l &= \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))}(1+\mathcal{O}(h)) \\ &\qquad + \Theta(h^{-2(ar+\eta(r))}) \\ &= \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r_{\max},r_{\max}}}} \frac{\omega_l}{(C+\epsilon)^{ar_{\max}+\eta(r_{\max})}} h_0^{-2(ar_{\max}+\eta(r_{\max}))} \\ &\qquad + \mathcal{O}(h^{1-2(ar_{\max}+\eta(r_{\max}))}).\end{aligned} \tag{1.93}$$

later, we denote the set Smooth $\cap\,\mathcal{L}_r$ as Smooth_r for simplicity. The overall accuracy estimation of the nonlinear weighting (1.84) is presented as

- $l \in$ Smooth

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'}\hat{\omega}_{l'}} = \frac{\omega_l}{\sum_{l'\in\text{Smooth}_r}\omega_{l'}} h^{2[a(r_{\max}-r)+\eta(r_{\max})-\eta(r)]}(1+\mathcal{O}(h)); \tag{1.94}$$

- $l \in$ Jump

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'}\hat{\omega}_{l'}} = \Theta(h^{2[a(r_{\max})+\eta(r_{\max})]}). \tag{1.95}$$

Now we can estimate the reconstruction error as

$$\begin{aligned}|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| &= |\sum_l \tilde{\omega}_l(u(\mathbf{x} - Q_l(\mathbf{x})))| \\ &\le \sum_{l\in\text{Smooth}} \tilde{\omega}_l|u(\mathbf{x}) - Q_l(\mathbf{x})| + \sum_{l\in\text{Jump}} \tilde{\omega}_l|u(\mathbf{x}) - Q_l(\mathbf{x})| \\ &\le \sum_r \sum_{\substack{l\in \\ \text{Smooth}\cap\mathcal{L}_{r,r}}} \Theta(h^{2[a(r_{\max}-r)+(\eta(r_{\max})-\eta(r))]})\mathcal{O}(h^r) \\ &\qquad + \sum_{l\in\text{Jump}} \Theta(h^{2(ar_{\max}+\eta(r_{\max}))})\mathcal{O}(1) \\ &= \mathcal{O}(h^{r_{\max}})\end{aligned} \tag{1.96}$$

Combining everything above and (1.81), we arrive at the formal error estimate of the reconstruction.

Lemma 1.4. *Let $\mathcal{R}(\mathbf{x})$ be the reconstruction. Assuming a qusi-uniform mesh, $h_0 = \Theta(h)$, there exists a constant $C > 0$ such that for any $\mathbf{x} \in E_0$,*

$$|u(\mathbf{x}) - \mathcal{R}(\mathbf{x})| \leq Ch^{r_{\max}}, \tag{1.97}$$

where $r_{\max}$ is defined by (1.86).

1.5.6 Handling Boundary Conditions

In our simulation, we couple FEM and FVM solver on a single mesh. Therefore, accurate and compatible boundary condition should be assigned accordingly. However, it is usually difficult to manage boundary condition for higher order finite volume and finite difference method, which require wide stencils. Some deal with the difficulty use ghost cells or points outside the boundary, and maintain the same stencil as if they were in the interior of the domain. However, the extrapolation or estimation of the manufactured values of ghost cells or ghost points are not straightforward. A notable method, the inverse Lax-Wendroff procedure, substitute the normal derivative of the solution with tangential and time derivatives using the PDE relation in an iteratvie way to impose more precise values for the ghost cells or points. The key to the above strategy is to have accurate and stable reconstruction near the boundary. ML-WENO gives the flexibility to add stenncils of the desired size that span back into the domain.

In order to assign proper boundary condition to a FVM discretization, we should determine whether the flow is inflow or outflow first. If $\mathbf{f}(u_B) \cdot \boldsymbol{\nu} > 0$, we recognize that it is a outflow boundary, therefore, no fixed boundary condition should be assigned. We use "free outflow" boundary that the numerical flux is simply $\mathbf{f}(u_B) \cdot \boldsymbol{\nu}$.

The inflow boundary can be understood in various ways. We can understand it as prescribing the inflow flux $\mathbf{f}_B$ or fixing the boundary value u_B as an equivelent to Dirichlet boundary in the context of Finite Element. In the special case when a zero flux is prescribed, we would call it noflow boundary. In the inflow case, there may be a discontinuous inflow shock wave. We require a swift response inside the domain to align with the specified Dirichlet value. We therefore add an extra constant stencil polynomial to the inflow boundary to accommodate the potential discontinuity.

There are some other special cases that require special treatment on the boundary, e.g., when symmetry should be enforced.

1.5.7 Numerical Test

For completeness, we test our ML-WENO algorithm with simple Riemann shock and rarefraction. Let $\mathbf{f}(u) = (u^2/2, u^2/2)$ be the Burgers flux function. In this example, we take the initial condition $u_0(x, y) = 1$, and the computational domain $\Omega = [0, 3] \times [0, 1]$. The true solution is

$$u(x,y,t) = \begin{cases} 0, & x < 0.5, x \geq 0.5t + 1.5, \\ (x-0.5)/t, & 0.5 \leq x < t + 0.5, \\ 1, & t + 0.5 \leq x < 0.5t + 1.5. \end{cases} \tag{1.98}$$

We forward martching to the time = 2.0, when trailing rarefaction reaches the leading shock. We use SSP2RK time stepping scheme.

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